How AI Reshapes Human Content Creation: The Case of Wikipedia*

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Abstract

We study how Grokipedia, an AI-generated encyclopedia, affects Wikipedia readership and editing. We compare covered biographies with uncovered biographies in the same field, matched on pre-launch activity and historical characteristics. Over the eight weeks after launch, difference-in-differences estimates imply increases in views and total edits equivalent to 16.2 and 46.5 percent of their pre-launch totals. Lower initial textual similarity predicts larger total-edit responses. Initially more dissimilar pages show more positive subsequent changes in similarity to Grokipedia's fixed launch-day text, especially where editing is greater. Contributors new to a page account for most of the participation increase, while highly experienced Wikipedians account for the largest share across experience groups. These findings support an inspiration channel: differences in AI-generated content can prompt contributors to revisit and revise Wikipedia entries. AI can therefore stimulate human knowledge production through content supplied on a competing platform.

JEL Classification: D83, L82, L86
Keywords: Wikipedia, Digital Platforms, Artificial Intelligence, User-Generated Content

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1 Introduction

Generative AI increasingly produces content that competes with human knowledge on separate platforms. Does this competition crowd out human contribution or prompt additional effort? We study this question through Grokipedia, an AI-generated encyclopedia launched as a rival to Wikipedia on October 27, 2025.[1] The AI-generated content appears on Grokipedia, while we measure readership, editing and textual changes on Wikipedia. Grokipedia thus provides an external source of competing information, rather than an AI assistant embedded in editors' work. Because language models draw training material from online content (Dodge et al., 2021), their effects on continued human contribution also matter for the knowledge resources on which future systems can draw.

Grokipedia may affect Wikipedia in opposing ways. Readers may turn to the rival, while publicity around its launch may increase interest in topics covered on both platforms. Differences in information or framing may also give Wikipedia editors ideas for additions or revisions. For example, Grokipedia's launch-day entry discussed anti-communism in Donald Trump's background, and on October 30 a Wikipedia editor added the category "American Anti-Communists" to his page.[2] The timing is suggestive, although we cannot observe whether the editor consulted Grokipedia. Editors may instead deliberately differentiate Wikipedia from its rival. In either case, editors must assess outside information against Wikipedia's sourcing and neutrality standards, and changes can be contested in discussion or reversed through review. Within these shared rules, individual volunteers decide whether and how to respond to Grokipedia. The net effects on readership, editing and content are therefore ambiguous ex ante and remain an empirical question.

We compare Wikipedia biographies covered by Grokipedia with uncovered biographies matched within five fields: Academia, Culture, Leaders, Politics and Sports. The underlying dataset contains approximately 1.4 million pages. Matching uses pre-launch activity

[1] Although Grokipedia's sustained visibility was limited, SimilarWeb estimates record about two million worldwide visits on October 28. The economically meaningful changes in Wikipedia activity documented below provide a revealed-preference argument for its relevance. Editors also explicitly discussed Grokipedia as a source of ideas on the archived Talk pages for Turkey (December 8–10, 2025) and RuneScape (November 13, 2025). These discussions concern possible improvements, not verified instances of adoption.

[2] Revision: `https://en.wikipedia.org/w/index.php?title=Donald_Trump&diff=prev&oldid=1319468163`.

histories and historical characteristics, yielding 80,909 treated pages. Recent weekly activity and trends satisfy common mean-balance benchmarks, and the pre-launch weekly comparisons show no systematic pre-trend. We estimate differences in differences using eight weeks before launch, September 1–October 26, and eight weeks afterward, October 27–December 21, 2025. Weekly comparisons trace the response.

Wikipedia activity increases relative to matched controls. Over the eight weeks after launch, the preferred estimates imply about 513 additional views and 0.296 additional total edits per page, equivalent to 16.2 and 46.5 percent of the treated pages' respective pre-launch totals. The estimated increases in views, total edits and minor edits are positive and statistically significant in all five fields. These findings are consistent with an AI-generated rival attracting additional attention and encouraging contributions to Wikipedia.

We next ask whether differences between the entries help explain the editing response. Lower initial textual similarity leads to larger total-edit responses, with a smaller effect for minor edits. We then compare weekly Wikipedia versions with each Grokipedia entry's fixed launch-day text. Pages with larger initial gaps show more positive subsequent changes in similarity, and this relationship is stronger on pages with more editing. The direction of textual change helps distinguish inspiration from deliberate differentiation, since either could generate more revisions where content initially differs. These patterns support an inspiration channel in which differences between entries prompt contributors to revisit a topic and revise its coverage.

Revision histories identify who contributes to the increase in participation relative to matched controls. Most comes from contributors new to a page in the observed revision history, and highly experienced Wikipedians account for the largest share across experience groups. Highly experienced editors making their first observed contribution to a page account for 38.7 percent of the total increase. The response therefore draws substantially on experienced Wikipedians becoming active on additional pages.

The setting connects two strands of work on generative AI. Studies of direct workplace assistance examine AI used within workers' production processes, including customer support (Brynjolfsson, Li, and Raymond, 2025). Studies of online knowledge communities

instead examine participation after ChatGPT becomes available. Burtch, Lee, and Chen (2024) study Stack Overflow and Reddit developer communities, finding declines on Stack Overflow but no evidence of declines on Reddit. del Rio-Chanona, Laurentsyeva, and Wachs (2024) compare Stack Overflow with other question-and-answer communities. Lyu et al. (2025) examine Wikipedia and show that post-ChatGPT changes vary with content overlap. In our setting, Grokipedia supplies an observable rival encyclopedia entry for the same individual, allowing us to connect initial textual distance, human participation, subsequent movement toward a fixed AI-generated text and contributor experience in one cross-platform setting.

The results suggest a source of complementarity beyond direct assistance: competing AI-generated content can draw attention to differences between entries and prompt experienced contributors to investigate them. The combination of editing, textual adjustment and editor composition is consistent with this interpretation. Wikipedia's sourcing and review procedures govern how information encountered elsewhere becomes accepted content.

Grokipedia differs from Wikipedia in several ways. In addition to being an AI-based information source, it is an alternative platform that competes for readers and a source based on alternative and unverified sources. And while we frame Wikipedia as a stock of information accumulated by individual editors, another view is that it is an institution that has a centralized response to Grokipedia.

We offer evidence that these alternative channels and framings are not too important here. If platform competition was the key mechanism, we would expect strategic differentiation of entries whereas we show there is convergence (as would happen under our view that Wikipedia editors use Grokipedia mainly as an inspiration). Our focus on AI effects is also consistent with other papers in the literature which focus on AI's impact on information creation on other platforms (Reddit, Stackoverflow, Pixiv). In terms of using alternative sources, our main results continue to hold when we flag references which are blacklisted on Wikipedia. Finally, we might think of Wikipedia as an institution/organization that is coordinating its response to Grokipedia. But Wikipedia is not a traditional institution: it is highly decentralized with editors acting autonomously

without any managerial guidance (and they work for free).

Section 2 reviews related work, and Section 3 describes the setting. Sections 4 and 5 present the model and data. Section 6 reports the empirical design and results, followed by the conclusion.

2 Literature

Our study contributes to research on generative AI, platform competition and voluntary knowledge production.

First, we contribute to the literature on how generative AI shapes user content creation. Empirical studies find that the availability of tools such as ChatGPT is associated with declines in human-authored posts and participation in online communities (Lyu et al., 2025; Burtch, Lee, and Chen, 2024; Glickman and Zhang, 2024; del Rio-Chanona, Laurentsyeva, and Wachs, 2024).[3] AI can also complement decentralized knowledge production directly. Evidence from neural machine translation across more than 100 Wikipedia language communities shows increased content creation without reduced quality or readership, with larger gains in better-resourced communities (Zhu and Walker, 2026). Evidence from Pixiv further shows that illustrators exposed to text-to-image AI competition reduce uploads, avoid tags favored by AI-generated content and broaden the intellectual properties they cover (Kim, Jin, and Lee, 2026). Differentiation across artworks is itself valuable on an art platform. Wikipedia and Grokipedia instead seek to document the same underlying subjects and facts, so movement toward the rival text can reflect information transmission. Our setting examines how contributors respond to AI content supplied on a competing platform.

We study a different channel: an AI-generated rival on a separate platform, whose content can redirect attention and reveal informational differences without providing Wikipedia contributors with a production tool. We find that this competitive entry

[3]Our results also relate to research on whether generative AI complements or substitutes for human skills (Brynjolfsson, Li, and Raymond, 2025; Acemoglu et al., 2022). Wikipedia contributors are unpaid and their output is freely available, so pro-social motives may shape their responses differently from those of paid workers. We also study knowledge creation rather than employment outcomes such as productivity or wages.

stimulates, rather than displaces, human contribution. This aligns with recent work on "human-in-the-loop" systems where AI serves as a "recombinant" tool, lowering the search costs for identifying knowledge gaps or framing new ideas (Agrawal, Gans, and Goldfarb, 2018). Our results suggest that AI-generated content serves as a reference point that highlights missing information or alternative perspectives. In other words, AI activates an "inspiration" channel: its novel framing and distinct informational content make gaps in the existing record salient, prompting editors to engage in expansionary and integrative work to close the distance between the platforms.[4]

Second, our findings connect to the literature on digital platform competition and strategic differentiation. Standard models of platform entry suggest that when a low-cost, automated entrant (Grokipedia) appears, the incumbent platform (Wikipedia) faces pressure to differentiate (Eisenmann, Parker, and Van Alstyne, 2011; Zhu and Iansiti, 2012). We study the response of Wikipedia's volunteer editors to this entry.

Our model emphasizes changes in reader attention and editors' responses to differences between entries. These channels can change contribution even when editing costs remain unchanged. This differs from the cost channel in Gans (2024), where AI reduces the cost of low-complexity tasks and shifts human effort toward harder problems through a free-rider threshold effect.

Third, we contribute to the literature on the private provision of public goods, particularly in online knowledge commons.[5] Prior work establishes that contributors to platforms like Wikipedia are motivated by peer recognition, social impact, and audience size (Zhang and Zhu, 2011; Gallus, 2017; Hinnosaar et al., 2022). Related field-experimental evidence from Zhihu distinguishes contribution-specific recognition from broader status signals: Zhang and Luo (2026) find that exogenously increasing peer upvotes raises subsequent answer volume, length, and quality, whereas adding followers has little overall effect. They interpret this contrast as consistent with peer recognition reinforcing intrinsic motivation, while follower counts function more as status signals. A key concern in this literature is

[4]Existing studies compare textual differences and citation practices between Grokipedia and the associated Wikipedia entry (Triedman and Mantzarlis, 2025; Yasseri, 2025). We examine changes in Wikipedia readership, editing and text following Grokipedia's entry.

[5]Some Wikipedia entries appear to be AI/LLM-generated (Brooks, Eggert, and Peskoff, 2024) despite Wikipedia policies prohibiting this practice. These are a small number of primarily recently created articles so it is not yet possible to measure their impact.

whether automated content generation will erode these intrinsic motivations by making human effort feel redundant.

Our findings add a role for competing AI-generated content in voluntary knowledge production. Editing responses are larger on pages with lower initial textual similarity to Grokipedia, and highly experienced contributors account for the largest share of the participation response across experience groups. External AI content can therefore encourage established volunteers to become active on additional pages. This finding connects voluntary contribution to research on how automation changes the tasks people perform (Autor, Levy, and Murnane, 2003).

3 Background on Wikipedia and Grokipedia

Wikipedia editors are responsible for verifiability, factual accuracy, reliable sourcing and neutrality, including when they use AI tools. Official guidelines allow AI-assisted drafting and editing but require substantive human review (Wikipedia contributors, 2024). The Wikimedia Foundation's AI strategy likewise treats generative models as tools to assist volunteer contributors (Owen, 2025).

Concerns about hallucinated citations, generic prose and unverifiable claims have led editors to strengthen safeguards against unreviewed AI content (Heater, 2025). In 2025, Wikipedia introduced a speedy deletion mechanism for articles suspected of being unvetted AI output (Hale, 2025). AI-assisted editing thus requires volunteers to check claims, repair citations and revise text to meet community standards. The Wikimedia Foundation also addressed the rise of generative AI encyclopedias in a November 10, 2025 blog post, emphasizing Wikipedia's value less than two weeks after Grokipedia's launch (Wikimedia Foundation, 2025b).

This policy environment makes Grokipedia a useful setting for studying the entry of an AI-native encyclopedia positioned as a competitor to Wikipedia, built in conjunction with xAI's "Grok" large language model.[6] Launched on October 27, 2025, its stated purpose is

[6]Grokipedia's announcement was largely a surprise. Elon Musk announced he would create an AI-based alternative to Wikipedia only three weeks before the roll-out, and the first public mention of this project occurred a month before that. Google Trends data for "Grokipedia" was at zero until the week of Elon Musk's announcement.

to provide a faster-moving, AI-supported alternative to Wikipedia. According to public descriptions, Grokipedia was initially populated by automatically ingesting large parts of English-language Wikipedia, followed by additional AI-generated editorial curation. The project aims to maintain an up-to-date, searchable knowledge base closely integrated with the Grok chatbot, and its early growth has been rapid: by November 21, 2025, Grokipedia already reported more than 1 million pages (compared with about 7.08 million articles on English-language Wikipedia when Grokipedia launched), although coverage remains uneven across topics.[7]

Wikipedia and Grokipedia compete for readers while using different production and governance systems. Both provide services with public-good characteristics and compete primarily for attention rather than price. Their entries are open to inspection, adaptation and criticism across platforms. Wikipedia relies on a resource-constrained volunteer labor force to create and revise articles. At launch, English Wikipedia reported 114,046 registered users active during the preceding 30 days.[8] A separate 2024 Wikimedia survey documents contributors across age groups and regions, although educational attainment is concentrated among degree holders: 81 percent reported post-secondary education and 42 percent a postgraduate degree (Wikimedia Foundation, 2025a). These cross-project survey respondents are not the same cohort as the English-Wikipedia activity count. Grokipedia instead uses a centralized, capital-intensive AI system to generate and curate entries, including material adapted from Wikipedia. During the study period, readers could suggest corrections but could not edit articles directly; Grok determined which suggestions to incorporate.[9] These differences allow competition to change where volunteers contribute as well as where readers seek information.[10]

[7]The historical Wikimedia statistics record reports 7,081,557 English Wikipedia articles at 21:45 UTC on October 27, 2025.

[8]The October 27, 2025 Wikimedia statistics record cited above reports active registered users, defined as users who performed an action in the preceding 30 days. This is an account-based activity measure, not a count of distinct people or an annual ten-edit threshold.

[9]See the contemporary account in Cruz (2025), which describes the suggestion workflow and Grok's control over updates.

[10]Wikipedia content is released under a Creative Commons Attribution-ShareAlike license, which permits extensive reuse and adaptation. Grokipedia's close textual overlap therefore does not raise standard copyright or fair-use constraints despite exceeding what would typically be permissible in proprietary markets. For imitation incentives, Wikipedia enforces community-governed sourcing and neutrality norms, including lists of non-authoritative sources that editors generally refuse to cite. Grokipedia does not publicly articulate comparable constraints. As a result, similarity reflects heterogeneous

Grokipedia's launch coincided with a sharp surge in user attention. Traffic estimates from SimilarWeb indicate that prior to launch, Grokipedia attracted only a few thousand global visits per day. Beginning on October 27, 2025, the day the platform went live, traffic spiked abruptly, reaching over 500,000 daily visits worldwide during its first week, with more than 100,000 daily visits originating from the United States alone. Peak attention occurred immediately after launch, exceeding two million global visits on October 28, before declining in the following days.

Grokipedia also attracted attention within the Wikimedia community. The March 10, 2026 Recent research column of Wikipedia's *Signpost* discussed studies comparing the two encyclopedias, including research on changes in Wikipedia editing following Grokipedia's entry (Bayer and Mitchsavl, 2026). This later coverage documents discussion among editors despite Grokipedia's limited general web traffic.

We focus on the first eight weeks after launch to study Wikipedia's initial response to Grokipedia.

4 A Stylized Model of Generative AI and UGC

We present a stylized model to clarify when generative AI acts as a substitute for, or a complement to, user-generated content (UGC), and to derive predictions for readership and editing.

Consider a Wikipedia page indexed by i. A unit mass of users consume information about topic i, and a fraction $\lambda \in (0, 1)$ of users are potential editors who may contribute to the Wikipedia page. Editors are intrinsically motivated and receive no monetary compensation.

An external AI platform (Grokipedia) may generate an entry for page i. Let $G_i \in \{0, 1\}$ indicate whether a Grokipedia entry exists. When $G_i = 1$, the AI entry may overlap with the existing Wikipedia content. Let $s_i \in [0, 1]$ denote the semantic similarity between the AI entry and the pre-existing Wikipedia article, where higher values correspond to greater overlap.

informational quality: overlap based on reputable sources may facilitate imitation, whereas overlap driven by partisan or non-authoritative outlets need not translate into editorial response.

4.1 User Attention

User traffic to Wikipedia depends on baseline informational value and on the presence of AI entries. Specifically, page views are given by

$$\text{Views}_i = \bar{V}_i + G_i(\theta - \delta s_i),$$

where $\bar{V}_i$ is baseline attention, $\theta \geq 0$ captures an attention-amplification effect from AI exposure, and $\delta \geq 0$ reflects substitution when the AI entry closely replicates Wikipedia content.

Lemma 1. ***(Average Effect of AI on Page Views)*** *Suppose $G_i = 1$ for treated pages and let $s_i \in [0, 1]$ denote the semantic similarity between the AI entry and the pre-existing Wikipedia article. Then the introduction of AI entries increases Wikipedia page views on average if*

$$\theta > \delta\, \mathbb{E}[s_i].$$

Moreover, the condition $\theta \geq \delta$ is a sufficient (but stronger) condition that guarantees page views weakly increase for all treated pages.

Lemma 1 distinguishes average effects from effects on individual pages. The parameter θ captures additional attention from greater awareness, lower search costs or users being directed toward Wikipedia. Substitution, δs_i, is stronger when the AI entry closely replicates existing content. Page views can therefore increase on average even if they decline on highly similar pages. The sufficient condition $\theta \geq \delta$ ensures that views weakly increase on every treated page, while $\theta > \delta\mathbb{E}[s_i]$ permits declines on some pages.

4.2 Editing Incentives

Each potential editor e faces an idiosyncratic editing cost c_{ie} drawn from a continuous distribution with CDF $F(\cdot)$ and density $f(\cdot) > 0$. Editors choose whether to contribute a discrete unit of effort.

The benefit from editing page i is

$$b_i = \kappa \cdot \text{Views}_i + G_i \cdot \rho(1 - s_i),$$

where $\kappa > 0$ captures pro-social benefits that increase with audience size (Zhang and Zhu, 2011), and $\rho > 0$ reflects an "inspiration" channel: when AI introduces novel framing or information, editors are more likely to respond.

Editor e contributes if $b_i \geq c_{ie}$. Aggregate editing activity for page i is therefore

$$E_i = F(b_i).$$

4.3 Implications and Testable Predictions

The model delivers three testable predictions that guide our empirical analysis.

Proposition 1. ***(Average Complementarity)*** *Suppose $\rho > 0$ and $\theta \geq \delta$. Then the introduction of a Grokipedia entry ($G_i = 1$) weakly increases Wikipedia editing activity for every treated page, and strictly increases expected editing activity on average relative to pages without a Grokipedia entry ($G_i = 0$).*

Proposition 1 shows how AI can complement human editing despite offering readers an alternative source. Under $\theta \geq \delta$, additional attention offsets substitution on every treated page. Differences in content or framing provide a further benefit to editors by drawing attention to possible revisions. Editing incentives therefore rise on pages with different content and weakly increase on the remaining pages.

Proposition 2. ***(Similarity Gradient Among Treated Pages)*** *Among Wikipedia pages that receive a Grokipedia entry ($G_i = 1$), aggregate editing activity is strictly decreasing in the semantic similarity between the Grokipedia entry and the pre-existing Wikipedia article:*

$$\frac{\partial E_i}{\partial s_i} < 0.$$

Proposition 2 predicts less editing when entries are more similar. A Grokipedia entry that repeats Wikipedia offers fewer prompts for revision than one that introduces different information or framing. Higher similarity reduces the inspiration benefit and increases substitution away from Wikipedia when $\delta > 0$; both effects lower editing incentives.

Proposition 3. ***(Irrelevance of Similarity for Control Pages)*** *For Wikipedia*

pages without a Grokipedia entry ($G_i = 0$), editing activity is independent of semantic similarity s_i.

Together, these predictions imply that similarity matters only within the treated sample, a feature that aligns directly with our empirical design. Formal proofs of all propositions are provided in Appendix A.[11]

5 Data

Our empirical analysis begins with the list of notable individuals compiled by Laouenan et al. (2022). Their cross-verified database contains 2.29 million individuals, approximately one-third of whom do not have an English Wikipedia page. We restrict the sample to individuals with an English-language Wikipedia entry in five primary occupational fields: Politics, Sports, Leaders, Academia and Culture. The resulting sample contains approximately 1.4 million individuals, about a fifth of all English Wikipedia articles at launch.

For these individuals, we combine Wikipedia API collections with Wikimedia pageview and MediaWiki history extracts. The activity panel covers September 1 through December 21, 2025 in complete Monday–Sunday weeks, providing eight observations before launch and eight afterward. Page views measure readership, while editing measures contributors' activity. The two outcomes distinguish whether the entrant attracts additional attention to Wikipedia from whether contributors respond by revisiting its content. Page views in this panel are user-classified dump views. The descriptive analysis of Grokipedia coverage uses all-agents Wikipedia API view measures.

We count total edits using matched records in the MediaWiki history extract and minor edits using its minor-edit flag. Minor edits typically include maintenance such as spelling, formatting and routine corrections. Because contributors supply the flag, it is an

[11] Appendix B extends the baseline model to allow Wikipedia editing to endogenously affect semantic similarity with Grokipedia and to capture editors' incentives to deliberately differentiate Wikipedia content in order to mitigate substitution. This extension provides a theoretical rationale for why the relationship between similarity and page views need not follow the simple substitution channel, as strategic differentiation can offset substitution effects. Allowing similarity to be endogenous does not alter any of the model's main predictions: Propositions 1, 2, and 3 continue to hold, and are formally shown to be invariant to this extension.

imperfect proxy for maintenance rather than a direct classification of substantive changes. We analyze the two editing outcomes separately to assess whether the response extends beyond activity marked as minor. We call the difference between total and minor edits non-minor editing; the data do not provide a separate major-edit flag. The extract does not impose a revision-creation-only restriction, so these outcomes should be interpreted as recorded editing activity. Historical Wikipedia article text and collected Grokipedia text provide the inputs for comparing the platforms' content.

In parallel, for every individual in the notable-people list, we check whether a corresponding Grokipedia page exists. We collect Grokipedia pages from its launch on October 27 through December 21, 2025, ending before a large late-December expansion.[12] Later additions complicate treatment classification by turning previously uncovered pages into covered pages; Section 6.1 explains how we restrict the control pool within our observation window. When one is present, we scrape the article text as well as any updates that have appeared during the empirical window. Applying this procedure to the 1.4 million notable individuals in our sample, we identify approximately 170,000 individuals with a Grokipedia entry, implying that about 12% of notable people with an English Wikipedia page in our focal fields were covered by Grokipedia during our data-collection window. This coverage, however, varies substantially across occupational domains: Grokipedia presence is highest in Culture (22.2%), followed by Leaders (12.1%), Politics (8.2%), Sports (8.2%), and Academia (6.6%). Combining all elements of the scraped data yields a unified person-level dataset that includes biographical identifiers, detailed Wikipedia traffic and edit histories, pre- and post-treatment Wikipedia content, and Grokipedia presence and text where available.

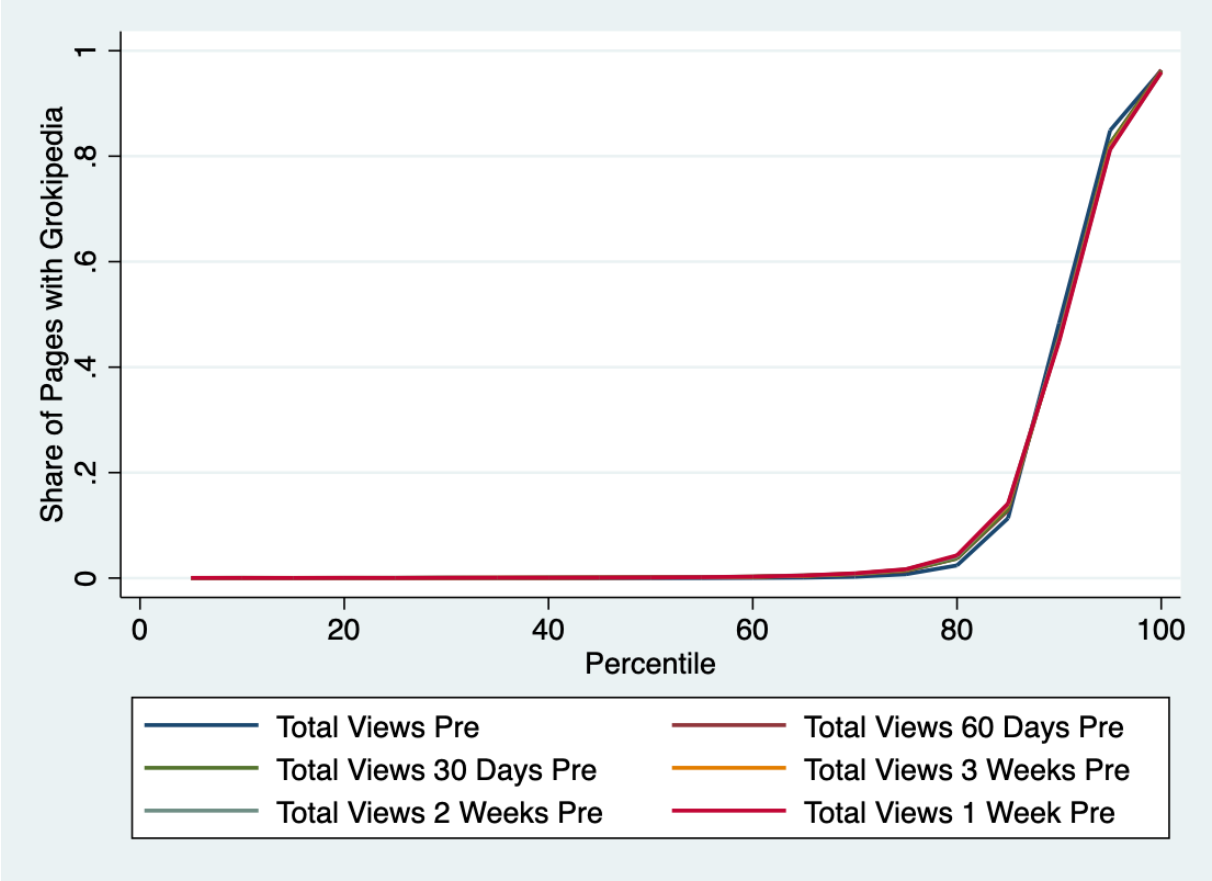
5.1 Which Pages Receive Grokipedia Entries?

Although Grokipedia's coverage rule is not publicly documented, pages that already attract attention may be especially valuable to an encyclopedia seeking an audience. The Grokipedia entries in our analysis have Wikipedia counterparts by construction, since

[12] Grokipedia reported roughly 890,000 articles early in the collection period and 1,016,241 by November 21. A December 30 report already put its count at 1,795,736, or about 1.8 million (Cruz, 2025).

our starting universe consists of existing Wikipedia biographies.[13] We therefore examine coverage across the pre-treatment view distribution. Figure 1 reports six API-based view measures: total pre-period views, 60-day and 30-day views, and the three individual weekly measures preceding launch. For each measure, we divide pages into 5-percentile bins and plot the average Grokipedia indicator within each bin. These descriptive measures characterize coverage, while the matching procedure below uses eight complete pre-treatment weeks.

Figure 1: Proportion of Grokipedia by 5-Percentile Bins of Pre-Treatment View Metrics



Coverage is concentrated toward the upper tail of the view distributions. This pattern makes an unconditional comparison between covered and uncovered pages uninformative about the launch effect: a popular biography can attract both AI coverage and subsequent Wikipedia attention. It also limits common support, since highly visible treated pages may

[13]Links use person identifiers and normalized page titles; identical URL slugs are not required. This sample construction does not establish that every Grokipedia entry had a Wikipedia analogue or that the platform created no new entries. Our analysis concerns activity on existing Wikipedia pages, not Wikipedia page creation.

have few comparable uncovered pages. Our matching procedure, discussed in Section 6.1, therefore restricts comparisons using pre-treatment activity and reports how many treated pages can be retained. It does not impose a fixed percentile band or automatically exclude every page above the 95th percentile.

We estimate linear probability models of Grokipedia coverage using log Wikipedia page views in the 60 days before launch, ln(Views 60 Pre). The specifications successively add prior editing activity, field indicators, historical biography word count, gender and age.

Pre-launch Wikipedia readership strongly predicts Grokipedia coverage. Across the four specifications in Table 1, the coefficient on log page views ranges from 0.145 to 0.151. A 10 percent increase in views corresponds to approximately a 1.4-percentage-point increase in the probability of coverage. Page views alone account for 47.7 percent of the variation in coverage. The coefficient remains similar after adding editing history, field indicators and biographical characteristics. Prior editing activity and historical biography length are also positively associated with coverage. Conditional on readership and editing, coverage remains higher in Culture, Leaders, Politics and Sports than in Academia.

These patterns motivate matching on pre-launch activity and guide the choice of matching covariates.

Table 1: OLS Regression on Grokipedia Presence

	Dependent variable: Grok indicator			
	(1)	(2)	(3)	(4)
ln(Views 60 Pre)	0.151***	0.147***	0.147***	0.145***
	(0.000133)	(0.000148)	(0.000152)	(0.000181)
ln(Edits 60 Pre +1)		0.0280***	0.0261***	0.0192***
		(0.000412)	(0.000412)	(0.000430)
Field: Culture			0.0665***	0.0667***
			(0.000743)	(0.000799)
Field: Leaders			0.0402***	0.0455***
			(0.000822)	(0.000896)
Field: Politics			0.0338***	0.0360***
			(0.000821)	(0.000880)
Field: Sports			0.0669***	0.0639***
			(0.000710)	(0.000764)
Wikipedia Word Count (10,000)				0.0309***
				(0.000325)
Gender: Male				-0.00540***
				(0.000570)
Gender: Other				0.0893***
				(0.0117)
Age				-0.0000341***
				(0.00000142)
Constant	-0.791***	-0.771***	-0.823***	-0.816***
	(0.000826)	(0.000872)	(0.00108)	(0.00132)
Observations	1422093	1422093	1422093	1291333
R^2	0.477	0.479	0.483	0.503

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

5.2 Similarity Between Grokipedia and Wikipedia Text

We compare each Grokipedia entry at its October 27, 2025 launch version with the corresponding Wikipedia article immediately beforehand, for treated pages where both texts are available. Stable person and page identifiers link biographies across platforms.[14] The histogram below and subsequent regressions use matched treated pages with nonmissing similarity scores. Matching compares pages on their pre-launch activity and historical characteristics, as described in Section 6.1. Neither textual similarity scores nor post-

[14]We thank `@lthlnkso` for generous assistance with the Grokipedia data.

launch activity enter the matching distance or restrictions.

Our hybrid similarity measure combines semantic overlap in meaning with lexical overlap in wording.

The first component uses the transformer-based embedding model all-MiniLM-L6-v2. It represents text as a vector and measures semantic overlap by the cosine between the two vectors. Texts can be close in this representation even when they describe the same events using different words or sentence structures. For example, two biographies may describe the same career trajectory while emphasizing different episodes or arranging them differently. This component is intended to capture overlap in meaning beyond exact wording.[15]

The second component uses TF–IDF cosine similarity to measure lexical overlap. The vectorizer is fitted to each pair of texts and weights words by their frequency and relative informativeness within that pair. Shared names, dates and terminology raise similarity, whereas wording changes can lower it even when the underlying facts are the same. This component is thus more sensitive to close textual reproduction than the semantic representation. Conversely, it can treat a paraphrase as less similar even when it conveys much the same information.

The two components respond to different aspects of encyclopedic content. For *Rodney (wrestler)*, the recorded TF–IDF score is 0.860 and the embedding score is 0.772. The texts share biographical material, names and wrestling terminology, while differing in presentation. The example illustrates why lexical and semantic measures need not agree.

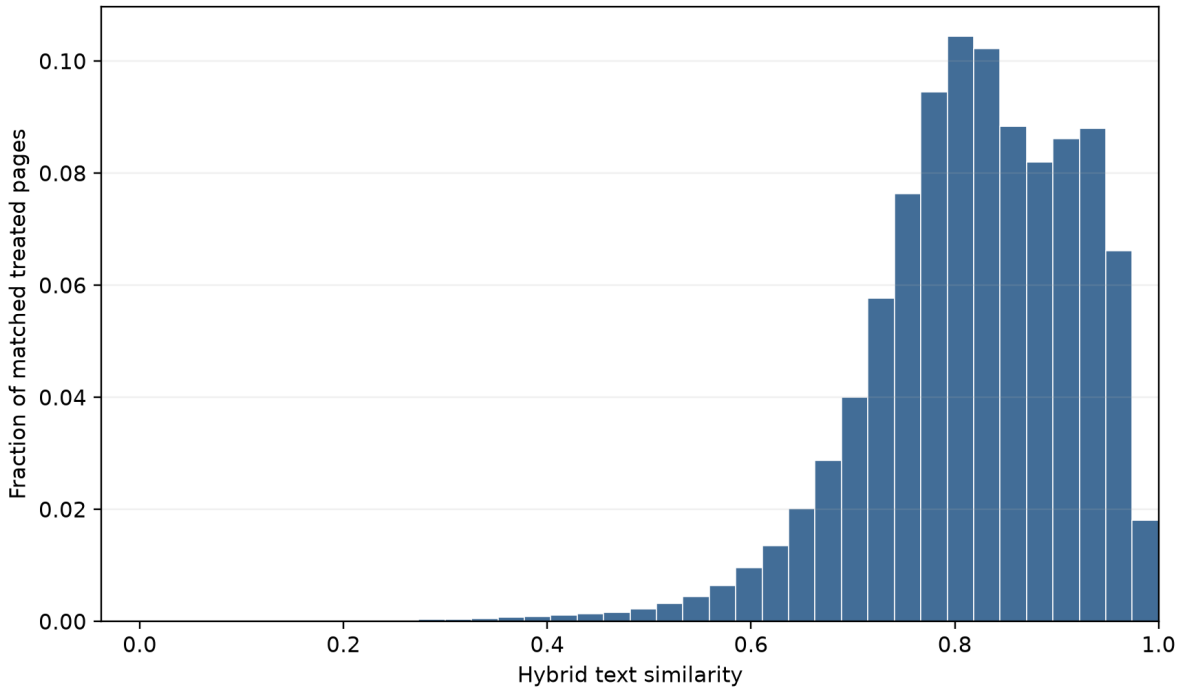
The hybrid score places weight 0.6 on embedding similarity and 0.4 on TF–IDF similarity. The greater embedding weight reflects our interest in content overlap, while the lexical component retains information about close wording. All reported regressions use this measure.

Figure 2 shows the distribution for the 80,687 matched treated pages with available similarity scores. Of these, 7.9% have scores above 0.95. Most have substantially lower scores, indicating differences from the pre-treatment Wikipedia text. This variation allows us to examine whether editing responses differ with the initial overlap between the two

[15]Each cleaned text is passed to the encoder once, using its first 256 wordpieces. The representation therefore need not cover every passage in a long article.

platforms' entries.

Figure 2: Similarity Scores in the Matched Treated Sample



6 Matching Estimation and Results

6.1 Matching

Grokipedia coverage is concentrated among pages with substantial prior attention, making the choice of a counterfactual central to the analysis. We construct matched pairs within each of the five occupational fields, so that an academic is compared with another academic and a politician with another politician. This restriction accounts for differences across fields in visibility and editing environments. Within each field, matching uses the eight complete Monday–Sunday weeks from September 1 through October 26, 2025. The post-treatment period begins on October 27 and ends on December 21, 2025, also spanning eight complete weeks.[16]

[16]The treatment indicator is a page-level classification of recorded Grokipedia presence. To reduce contamination, we remove nominally uncovered pages from the control pool if Grokipedia metadata indicate that they received an entry between November 1 and December 21.

The preferred matching vector is guided by the coverage estimates in Section 5.1 and uses both weekly levels and trends in Wikipedia attention and editing before launch.[17] It includes $\log(1 + \text{views})$ and $\log(1 + \text{total edits})$ in each of the eight pre-treatment weeks, the logarithm of one plus each outcome's eight-week mean, an indicator for any editing during those weeks, and raw linear weekly slopes for views and total edits. Weekly values distinguish pages with different activity paths even when their mean levels or trends are similar.

We also include three historical covariates in their original units: pre-launch 60-day editing activity, historical biography word count, and average annual Wikipedia readership across editions in 2015–2018 from Laouenan et al. (2022). These characteristics capture prior editorial investment, the accumulated stock of content and persistent popularity beyond the immediate eight-week window. Matching on both recent dynamics and historical characteristics seeks controls with a comparable underlying propensity to attract readers and contributors. Only pre-treatment variables enter the matching distance and restrictions; post-launch outcomes and numerical similarity scores are not matching coordinates.

We use nearest-neighbor multivariate covariate matching with a standardized distance, combined with exact restrictions and a caliper, conventional ingredients in matched observational designs (Stuart, 2010). Combining multivariate matching with a caliper also has a long precedent (Rosenbaum and Rubin, 1985); here the caliper is on readership, and the metric is weighted standardized Euclidean distance. Within field f, the distance between treated page i and control j is

$$D_f(i,j) = \left[\sum_{k=1}^{24} w_k \left(\frac{X_{ik} - X_{jk}}{s_{fk}}\right)^2\right]^{1/2},$$

where X contains the 21 recent-activity coordinates and three historical covariates, w_k is the weight for each of the covariates,[18] and s_{fk} is the sample standard deviation among

[17] In Table 1, adding editing history, fields and biographical characteristics raises R^2 from 0.477 to 0.503. The modest incremental explanatory power supports emphasizing pre-launch activity, although it does not establish that additional covariates could not improve balance.

[18] The weights are not derived from a theoretical optimum or an external prespecification. The final specification was adopted after earlier estimates were available, with these weights carried over unchanged. Under the exact edit-profile restriction below, the edit-related coordinates have zero within-pair differ-

all eligible treated and control pages in that field.

A control must have exactly the same total-edit count as its treated page in each of the eight pre-treatment weeks. This restriction matches the complete weekly edit profile, rather than only the eight-week total. We also impose a 0.10 caliper on the logarithm of one plus mean weekly views. We select one control per treated page, allowing at most ten uses of a control.[19]

Of 174,763 eligible treated pages, 80,909 are matched, a retention rate of 46.3%.[20] They use 24,670 distinct controls. Unmatched pages lack an acceptable candidate under the search, activity restrictions, calipers and reuse rule. The resulting estimates therefore apply to treated pages with acceptable matches rather than to the entire universe of Grokipedia entries. Figure 3 displays the eight weekly pre-treatment view and minor-edit gaps, standardized within field. Appendix C provides the eight weekly outcome balances and historical-covariate diagnostics. Appendix D adds exact primary-citizenship matching and birth year as a distance coordinate.

The largest absolute standardized weekly outcome gap across fields is 0.043, and the largest standardized slope gap is 0.025. Both are well below 0.1, a commonly used benchmark for good mean balance (Austin, 2009).[21] Views and minor edits retain some pre-launch differences, while total-edit differences are zero by construction. The latter is an imposed restriction, not independent evidence of parallel trends. Historical characteristics are not matched exactly: the largest absolute standardized gap is 0.469 for biography length in Academia (Appendix C). Thus the favorable recent-activity balance does not extend to every historical covariate. Figure 4 displays the raw weekly treated-control gaps.

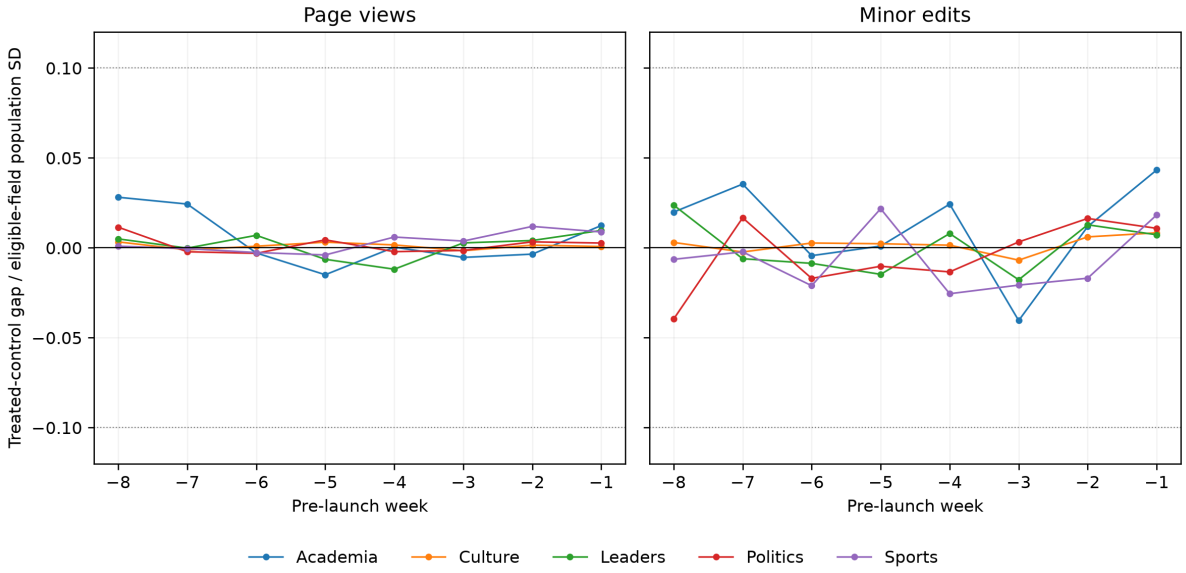
ences, so their weights do not affect the ranking of eligible controls.

[19] Within each field and exact edit-profile group, the implementation examines the 250 nearest candidates, or all candidates if fewer are available. It applies the view caliper and reuse cap in frozen input order, retaining the original nearest-neighbor tie procedure. The weekly edit, mean-edit, any-edit and edit-slope coordinates remain in the distance, although the exact-profile restriction sets their within-pair differences to zero.

[20] The 93,854 unmatched treated pages face limited common support under the matching rules: 23,136 have no control with the same weekly edit profile, 44,743 have no view-caliper candidate among the nearest 250 controls, and 25,975 exhaust the reuse cap within that candidate set. Unmatched treated pages have median pre-launch readership of 1,038 views per week, compared with 242 among matched pages (means: 3,463 and 397). Grokipedia's concentration on high-view biographies thus leaves fewer comparable uncovered pages once the activity restrictions are imposed.

[21] The reported gaps use the complete eligible field population's standard deviation. Using instead the pooled standard deviation of treated and matched-control groups, as in Austin (2009), the corresponding maxima are 0.091 and 0.062, also below 0.1. The benchmark is a rule of thumb, not a universal criterion.

Figure 3: Weekly Pre-Treatment Balance in Views and Minor Edits. Gaps are treated minus matched-control means, divided by the eligible-field population standard deviation of the corresponding weekly outcome. Controls are weighted by their use in matched pairs. Total edits are omitted because their weekly balance is imposed exactly by the matching restriction.



The symmetric eight-pre/eight-post comparison observes both groups over the same calendar weeks and differences out persistent matched-pair level differences.

Identification relies on the treated-control gap remaining stable absent Grokipedia. Matching addresses observed activity histories and historical characteristics, but differential unobserved shocks could still affect the comparison. The weekly pre-launch gaps in Figure 4 provide a diagnostic for this assumption.

6.2 Effects on Views and Edits

Let $j(i)$ denote the matched control for treated page i, and let $\bar{Y}_i^{\text{pre}}$ and $\bar{Y}_i^{\text{post}}$ be its average weekly outcome in the eight pre- and eight post-launch weeks. The preferred contrast is

$$\widehat{\tau}_{8,8} = \frac{1}{N_T} \sum_{i:G_i=1} \left[(\bar{Y}_i^{\text{post}} - \bar{Y}_{j(i)}^{\text{post}}) - (\bar{Y}_i^{\text{pre}} - \bar{Y}_{j(i)}^{\text{pre}}) \right].$$

The tables report $8\widehat{\tau}_{8,8}$ and its standard error. These total-count units compare two equally long eight-week periods. Dividing both coefficients and standard errors by eight gives the weekly-average contrast. The individual-level DiD outcomes used below have

the same total-count units. Standard errors equal the sample standard deviation of pair differences divided by $\sqrt{N_T}$.

Table 2: Pooled Eight-Week DiD Estimates

	Views	Total edits	Minor edits
DiD	512.94***	0.296***	0.092***
	(33.03)	(0.018)	(0.005)
Matched pairs	80,909	80,909	80,909

Standard errors in parentheses. Matching uses eight pre-treatment weeks, September 1–October 26, 2025, and the three historical covariates. Post-treatment weeks cover October 27–December 21, 2025. DiD coefficients and standard errors are reported in eight-week total-count units: divide both by eight for the difference between average weekly post- and pre-treatment gaps. One control per treated page, with at most ten uses per control. DiD standard errors use pair-level differences and are not adjusted for reused controls. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$. Matching uses weighted standardized Euclidean distance, exact eight-week total-edit profiles within field, and a 0.10 log-mean-view caliper.

Table 2 reports pooled increases of 512.94 views, 0.296 total edits and 0.092 minor edits per page over eight weeks, using 80,909 matched pairs. The treated pages' pre-launch means are 3,172.38 views, 0.637 total edits and 0.187 minor edits. The estimates therefore equal 16.2%, 46.5% and 49.3% of the respective baselines, with treated pages weighted equally across the pooled sample.

All three pooled estimates are positive and statistically significant at the 1 percent level. The view increase is consistent with the attention channel in Lemma 1: the AI entrant may direct readers toward Wikipedia as well as offer an alternative. The edit increase is consistent with Proposition 1, under which attention and differences in content raise the return to contributing. Total edits rise by far more than minor edits, indicating that the response extends beyond activity carrying the minor flag. These patterns support short-run complementarity.

Table 3 shows positive estimates for views, total edits and minor edits in all five fields, each statistically significant at the 1 percent level. View effects range from 328.99 in Politics to 680.22 in Leaders. Total-edit effects range from 0.243 in Politics to 0.430 in Leaders, while minor-edit effects range from 0.068 in Culture to 0.131 in Academia.

Increased attention and non-minor editing occur across all five fields. Differences in magnitude also reflect differences in baseline activity and in the pages with suitable controls.

Table 3: Eight-Week DiD Estimates by Field

Field	Views	Total edits	Minor edits	Matched pairs
Academia	387.98***	0.403***	0.131***	4,808
	(57.15)	(0.074)	(0.023)	
Culture	473.08***	0.270***	0.068***	32,690
	(53.93)	(0.027)	(0.007)	
Leaders	680.22***	0.430***	0.114***	12,880
	(105.36)	(0.050)	(0.013)	
Politics	328.99***	0.243***	0.121***	8,461
	(91.90)	(0.062)	(0.016)	
Sports	572.10***	0.254***	0.095***	22,070
	(55.69)	(0.035)	(0.009)	

Standard errors in parentheses. Matching uses eight pre-treatment weeks, September 1–October 26, 2025, and the three historical covariates. Post-treatment weeks cover October 27–December 21, 2025. DiD coefficients and standard errors are reported in eight-week total-count units: divide both by eight for the difference between average weekly post- and pre-treatment gaps. One control per treated page, with at most ten uses per control. DiD standard errors use pair-level differences and are not adjusted for reused controls. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$. Matching uses weighted standardized Euclidean distance, exact eight-week total-edit profiles within field, and a 0.10 log-mean-view caliper.

6.3 Weekly Patterns Around the Launch

Figure 4 displays weekly treated-control differences for the baseline matched pairs. For each week t, it reports

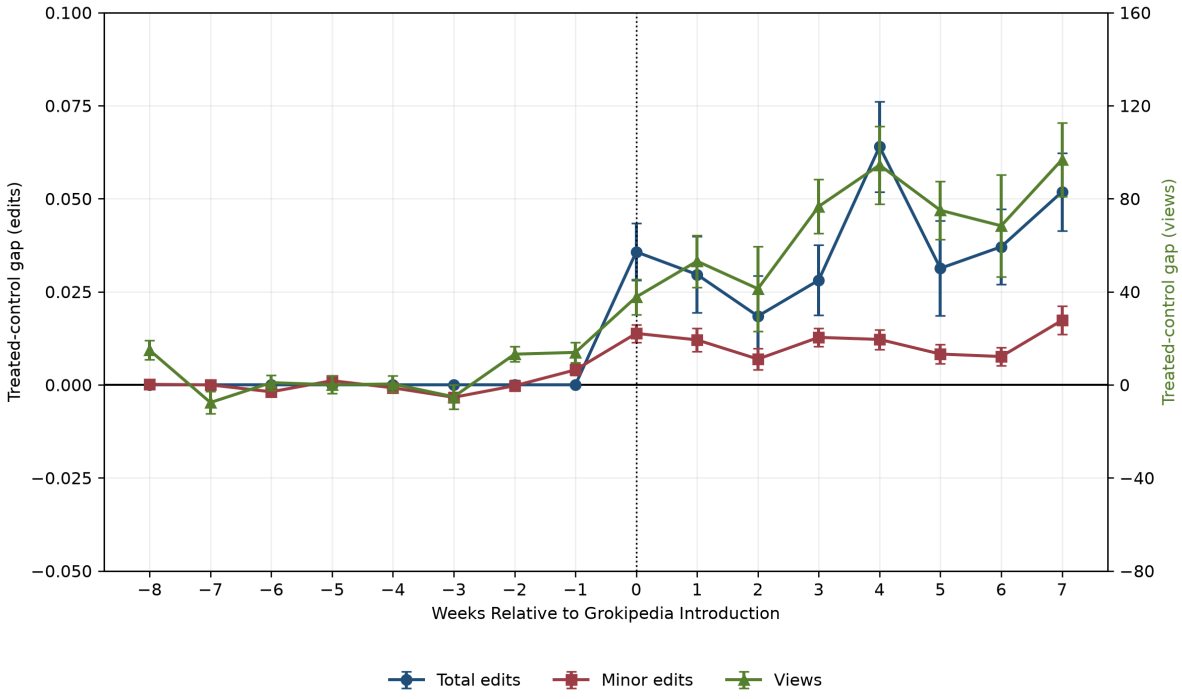
$$\widehat{\mathrm{ATT}}_t = \frac{1}{N_T} \sum_{i:G_i=1} (Y_{it} - Y_{j(i)t}).$$

These are raw weekly treated-control differences, not coefficients normalized to an omitted pre-treatment period. The figure shows all eight pre-treatment weeks and all eight post-treatment weeks with 95% confidence intervals. Its post-period average minus its pre-period average equals $\widehat{\tau}_{8,8}$.

Pre-launch total-edit gaps are exactly zero because matching requires identical weekly edit profiles. View and minor-edit gaps vary across the pre-period, and some are statistically distinguishable from zero. After launch, every weekly pooled gap is positive for all three outcomes, with its 95% confidence interval above zero. The positive gaps persist throughout the eight post-launch weeks. The panel estimates are therefore not driven by a one-time spike that quickly reverts to zero.

Total editing activity exceeds activity marked as minor in the post-launch weeks. Some of the response may therefore involve maintenance, but the increase extends beyond

Figure 4: Weekly Treated-Control Gaps Around Grokipedia's Launch. Weeks −8 to −1 cover September 1–October 26, 2025. Weeks 0 to 7 cover October 27–December 21, 2025. The left axis shows total and minor edits, and the right axis shows views. The zero lines are aligned. The 95% intervals use the pair-level standard errors described in the text.



that margin. Alongside the increase in views, this pattern is consistent with Grokipedia drawing attention to existing pages and prompting contributors to revisit them.

6.4 Textual Similarity and Wikipedia Revisions

The launch anecdote in the Introduction illustrates a possible connection between AI-generated framing and Wikipedia revisions. Proposition 2 formalizes this idea: an entry that largely repeats Wikipedia may reveal little that editors did not already know, whereas differences can make omissions or contested claims more salient. This predicts a larger editorial response where textual overlap is lower. Similarity is a proxy for that contrast, not a direct measure of how much useful new information an AI entry contains.

To examine this prediction, we use each treated page's total-count DiD outcome,

$$\Delta Y_i^{\text{DiD}} = 8\left[(\bar{Y}_i^{\text{post}} - \bar{Y}_{j(i)}^{\text{post}}) - (\bar{Y}_i^{\text{pre}} - \bar{Y}_{j(i)}^{\text{pre}})\right],$$

and estimate

$$\Delta Y_i^{\text{DiD}} = \alpha + \beta\,\text{Similarity}_i + \mathbf{X}_i'\boldsymbol{\gamma} + \varepsilon_i.$$

We first include similarity alone, then add field indicators, and finally historical word count, gender and the individual's age. Field indicators account for average differences across subject areas, while the page controls account for observed differences in content and biographical characteristics. All specifications use the baseline assignments and the same eight-plus-eight-week contrast, with conventional OLS standard errors.

Table 4: Similarity and Eight-Week Matched-Pair DiD Changes in Editing

	Total (1)	Total (2)	Total (3)	Minor (1)	Minor (2)	Minor (3)
Similarity score	-0.533***	-0.541***	-0.544***	-0.113**	-0.104**	-0.109**
	(0.178)	(0.180)	(0.188)	(0.047)	(0.047)	(0.050)
Field indicators	No	Yes	Yes	No	Yes	Yes
Page controls	No	No	Yes	No	No	Yes
Observations	80,687	80,687	75,450	80,687	80,687	75,450
R^2	0.000	0.000	0.003	0.000	0.000	0.003

Standard errors in parentheses. Outcomes are individual matched-pair DiD changes in eight-week total-count units. Both periods contain eight weeks: September 1–October 26 and October 27–December 21, 2025. Conventional OLS inference. Page controls are historical word count, gender and age. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Lower initial similarity is associated with larger total-edit responses across all three specifications in Table 4. With full controls, a 0.10 reduction in similarity is associated with 0.054 additional total edits over eight weeks, equivalent to 8.5% of the pooled treated pages' pre-launch mean. The association is also negative for minor edits, although its magnitude is smaller. The total-edit coefficients are significant at the 1 percent level and the minor-edit coefficients at the 5 percent level in all three specifications. The point estimates imply that the similarity gradient is concentrated in non-minor edits.

The corresponding association with views is positive but imprecisely estimated (Appendix E).

The negative total-edit and minor-edit associations are consistent with the inspiration mechanism in Proposition 2, conditional on the observed controls.[22]

We next examine whether Wikipedia revisions are associated with movement toward Grokipedia's text. This distinguishes inspiration from a strategic-differentiation response to competition for readers: deliberate differentiation would push the entries apart, as in Appendix B, whereas movement toward the rival text is consistent with information transmission. We hold each Grokipedia page fixed at its October 27, 2025 version and reconstruct Wikipedia immediately before launch and at the end of each of the eight post-launch weeks, through December 21. We apply the same cleaning rules and hybrid similarity calculation at every date. The balanced panel contains 80,653 matched treated pages with nine usable snapshots.

Let S_{i0} denote this reconstructed pre-launch similarity and S_{i8} its week-eight value. The change $\Delta S_i = S_{i8} - S_{i0}$ is positive when Wikipedia moves closer to the fixed Grokipedia text. Define the initial textual gap as $g_i = 1 - S_{i0}$. We estimate the page-level relationship

$$\Delta S_i = \alpha + \beta g_i + \mathbf{X}_i'\boldsymbol{\gamma} + \varepsilon_i,$$

adding field indicators and the established page controls in sequence. The initial-gap coefficient is positive and significant at the 1 percent level in all three specifications in Table 5. With full controls, the coefficient is 0.006362 (standard error 0.000508), using

[22]The informational value of a textual difference also matters. Verifiable additions may help editors identify omissions, while unreliable claims may prompt correction or no response. The similarity score does not distinguish these cases.

75,421 complete cases. Pages with larger initial gaps thus exhibit more positive subsequent similarity changes, conditional on these characteristics.[23] The initial-gap coefficient is also positive within each occupational field, with magnitudes reported in Appendix F.

Table 5: Text Convergence — Initial Textual Gap, Editing and Eight-Week Change in Similarity

	(1)	(2)	(3)	(4)
Initial gap	0.006856***	0.006595***	0.006362***	-0.003796***
	(0.000499)	(0.000507)	(0.000508)	(0.000604)
log(1 + PostTotalEdits)				-0.005337***
				(0.000360)
Initial gap ×				
log(1 + PostTotalEdits)				0.020211***
				(0.001709)
Field indicators	No	Yes	Yes	Yes
Page controls	No	No	Yes	Yes
Observations	80,653	80,653	75,421	75,421
R^2	0.004	0.005	0.005	0.030

Standard errors in parentheses. HC1 robust inference uses residual-degrees-of-freedom t statistics. The outcome is week-eight similarity minus pre-launch similarity, holding Grokipedia text fixed at October 27, 2025. Initial gap equals one minus pre-launch Wikipedia–Grokipedia similarity. Positive gap coefficients in columns (1)–(3) indicate more positive change among initially more distant pages. A positive interaction in column (4) means this gradient becomes more positive with editing. PostTotalEdits sums recorded activity over October 27–December 21. Column (4) includes both main effects and their interaction. Page controls are historical word count, gender and age. These are descriptive associations. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Over eight weeks, the median page contains 657 cleaned words, receives 0 recorded edits and has 0 actual revisions. The shares with no recorded edits and no actual revisions are 51.11% and 52.87%, respectively, and 80.85% have unchanged endpoint similarity within numerical tolerance. Among pages with positive observed textual turnover, median gross turnover is 1.42% of baseline length.[24]

Among the 4,698 pages in the lowest initial-similarity quartile whose text changes, the

[23] The regression remains descriptive. Initial gap is constructed from baseline similarity, which also enters subsequent change with a minus sign. Baseline measurement error can therefore mechanically induce a positive relationship. Using initial gap does not eliminate mathematical coupling or regression to the mean. Inherited cleaning rules can also amplify formatting changes.

[24] Gross turnover sums token additions and removals between adjacent weekly snapshots, divided by baseline cleaned length. Replacements count twice and repeated changes can be counted repeatedly, so it is not the share of unique words changed. "Words" are whitespace-delimited cleaned tokens. The hybrid index is not additive across words: its embedding component uses the first 256 wordpieces, while TF–IDF uses the full cleaned text. Turnover therefore supplies context, not a per-word conversion of similarity.

mean proportional reduction in the initial gap, $\Delta S_i/(1 - S_{i0})$, is 0.429% (95% confidence interval: 0.220–0.638%).[25] Across the full sample, mean similarity changes from 0.819033 to 0.818512.

Column (4) asks whether the convergence gradient varies with post-launch editing intensity. Let $e_i = \log(1 + \text{PostTotalEdits}_i)$, where post-launch edits sum recorded total activity over the same eight weeks. We estimate

$$\Delta S_i = \alpha + \beta_1 g_i + \beta_2 e_i + \beta_3 (g_i e_i) + \mathbf{X}_i' \boldsymbol{\gamma} + \varepsilon_i.$$

Column (4) includes field indicators and the same page controls as column (3). The initial-gap-by-editing interaction is 0.020211 (standard error 0.001709), significant at the 1 percent level. The relationship between the initial gap and subsequent similarity change becomes more positive on pages with more post-launch edits.

At an initial similarity of 0.60 and 2 post-launch edits, the median positive count, the fitted change corresponds to closure of 0.614% of the initial gap (95% confidence interval: 0.458–0.769%).[26]

Lower textual overlap is associated with a larger total-edit response and more positive subsequent changes in similarity to Grokipedia's fixed launch-day text, especially on pages with more editing. These patterns are consistent with an inspiration channel in which differences in content or framing prompt contributors to revise Wikipedia.

6.5 Editor Participation

The increase in editing relative to matched controls raises a further question: does it reflect greater participation by incumbent page editors, inexperienced contributors, or experienced editors entering other pages? MediaWiki history identifies the contributor associated with each revision. We exclude revisions flagged as bots by historical group

[25] Quartiles are fixed at baseline, and proportional changes are calculated among token-changing pages below the predefined 0.99 baseline-similarity ceiling. The measure is a fraction of the initial gap rather than a percentage-point change in the similarity score.

[26] We average the fitted prediction over observed field and page controls among complete cases with positive cleaned length and baseline similarity within 0.025 of 0.60. Support requires at least 50 such pages and an edit count between their observed 1st and 99th percentiles. The confidence interval uses the HC1 covariance of the standardized prediction. The global linear projection does not constrain fitted change to zero at zero edits.

metadata and count each identifiable editor once per page-week, irrespective of how many revisions they make. The outcome therefore measures editor participation rather than revision volume. We retain 80,788 of the original 80,909 matched pairs for which both pages can be linked to stable revision-history page IDs. The remaining 121 pairs have unresolved identities, comprising one treated-page link and 120 control-page links, and are excluded without rematching.

We distinguish page familiarity from Wikipedia experience. *Familiarity* is based on whether an editor is contributing to a given page for the first observed time or returning to it in later active weeks. This classification uses all available same-page revision history since July 2015. *Experience* instead counts prior English-Wikipedia revisions across all namespaces, frozen strictly before September 1, 2025. We distinguish low experience (0–9 revisions), medium experience (10–999), high experience (1,000 or more), and unknown experience, including IP and temporary identities.[27] All participation is classified as either first-time or returning, and separately into one of the four experience categories.

Table 6 compares the eight-week sums of weekly editor counts before and after launch, using September 1–October 26 and October 27–December 21. An editor active in several weeks contributes several participations. The total DiD is 0.180 editor-week participations per treated page relative to its control (standard error 0.019). Participation increases on both treated and control pages, with a larger increase on treated pages. Panel A attributes 0.132 participations, or 73.5% of the relative response, to first-time-on-page editors. Returning editors account for 0.048 participations.

Panel B shows that high-experience editors contribute 0.115 participations, accounting for 64.2% of the total DiD. Low- and medium-experience editors account for 5.3% and 7.2%, respectively, while the unknown group contributes 23.3%. The largest joint category consists of highly experienced editors contributing to a page for the first observed time: 0.070 additional editor-week participations per page (standard error 0.008), or 38.7% of the total increase. This category is estimated directly from the joint classification. The response thus draws substantially on experienced Wikipedians becoming active on

[27]First-time means first observed in the available history, which is left-censored before July 2015. Observable editor identities need not correspond one-to-one to individuals. Temporary-account adoption during the post period can also create apparent entry, so such accounts remain in the unknown-experience group rather than being classified as novices.

Table 6: Eight-Week DiD in Wikipedia Editor Participation by Editor Type

	Eight-week DiD	Share of total DiD
Total editor participation	0.1797***	100.0%
	(0.0187)	
Panel A. Page familiarity		
First-time on page	0.1321***	73.5%
	(0.0151)	
Returning to page	0.0476***	26.5%
	(0.0067)	
Panel B. Prior English-Wikipedia experience		
Low: 0–9 prior revisions	0.0095***	5.3%
	(0.0027)	
Medium: 10–999	0.0130***	7.2%
	(0.0036)	
High: 1,000+	0.1154***	64.2%
	(0.0106)	
Unknown experience	0.0419***	23.3%
	(0.0093)	
Matched pairs	80,788	

Standard errors in parentheses. Inference clusters by matched control page to allow for control reuse. Outcomes are eight-week sums of weekly unique identifiable non-bot editor counts, not distinct editors over the entire period. DiD compares September 1–October 26 with October 27–December 21, 2025. First-time means the first observed week on that page using histories since July 2015. Experience is cumulative English-Wikipedia revisions before September 1, fixed throughout both periods. IP and temporary identities have unknown experience. All retained matched pairs have resolved page IDs and processed histories on both sides. Panels A and B separately exhaust the total. Shares use unrounded point estimates. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

additional pages.

Prior expertise may matter differently when AI supplies an external source to evaluate than when it directly assists a task. In customer support, AI assistance produces especially large productivity gains among novice and lower-skilled workers (Brynjolfsson, Li, and Raymond, 2025). A three-month patent-lawyer experiment similarly finds larger assisted-drafting gains among juniors, but subsequent gains on an unaided judgment task are concentrated among seniors (Autor et al., 2026). These results suggest that expertise can shape how people respond to AI-generated information. We contribute to this ongoing literature in another environment: volunteer participation in response to a rival encyclopedia.

Experienced editors may face lower costs of checking unfamiliar claims and reconciling them with Wikipedia's sourcing and neutrality standards. An alternative encyclopedia entry could consequently draw their attention to pages where that expertise is useful.

7 Conclusion

Grokipedia's launch increased readership and editing on covered Wikipedia pages relative to matched controls. Over the eight weeks after launch, the estimated increases amount to 16.2 percent of pre-launch views and 46.5 percent of pre-launch total edits. Pages with less initial textual overlap have larger total-edit responses, with the similarity gradient concentrated in non-minor edits. Initially more dissimilar Wikipedia pages also move relatively closer to Grokipedia's fixed launch-day text, particularly where editing is more intensive. This relative convergence weighs against strategic differentiation as the primary response to the rival platform and supports the inspiration channel.

Most of the increase in participation comes from contributors new to a page in the observed revision history. Highly experienced Wikipedians account for the largest share across experience groups, indicating that the response draws substantially on established contributors becoming active on additional pages.

Future work could examine how these responses develop over longer periods, affect the quality and neutrality of Wikipedia content (Greenstein and Zhu, 2012), and shape public trust in Wikipedia amid AI competition (Hinnosaar et al., 2023).

These findings suggest that competing AI-generated content can encourage human contribution by drawing attention to differences in coverage or framing. Contributors decide which pages to edit and how to revise them, while Wikipedia's sourcing and review procedures govern the resulting content. AI can therefore stimulate voluntary knowledge production through content supplied on a competing platform.

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Appendix A Proofs for Section 4

We provide formal proofs for the three propositions stated in Section 4.

Appendix A.1 Setup

Aggregate editing activity on page i is given by

$$E_i = F(b_i),$$

where $F(\cdot)$ is a continuous cumulative distribution function with strictly positive density $f(\cdot)$, and

$$b_i = \kappa \cdot \text{Views}_i + G_i \cdot \rho(1 - s_i).$$

Page views satisfy

$$\text{Views}_i = \bar{V}_i + G_i(\theta - \delta s_i),$$

with parameters $\kappa, \rho, \theta, \delta > 0$.

Appendix A.2 Proof of Proposition 1 (Average Complementarity)

Aggregate editing activity on page i is given by $E_i = F(b_i)$, where $F(\cdot)$ is a strictly increasing cumulative distribution function and

$$b_i = \kappa \cdot \text{Views}_i + G_i \cdot \rho(1 - s_i).$$

Recall that page views satisfy

$$\text{Views}_i = \bar{V}_i + G_i(\theta - \delta s_i).$$

For a treated page ($G_i = 1$), aggregate editing activity is therefore

$$E_i(1) = F\Big(\kappa(\bar{V}_i + \theta - \delta s_i) + \rho(1 - s_i)\Big),$$

while for an untreated page ($G_i = 0$),

$$E_i(0) = F(\kappa \bar{V}_i).$$

The difference in editing activity between treated and untreated pages is

$$E_i(1) - E_i(0) = F\Big(\kappa(\bar{V}_i + \theta - \delta s_i) + \rho(1 - s_i)\Big) - F(\kappa \bar{V}_i).$$

Because $F(\cdot)$ is strictly increasing, a sufficient condition for $E_i(1) > E_i(0)$ is that

$$\kappa(\theta - \delta s_i) + \rho(1 - s_i) \geq 0.$$

Since $s_i \in [0, 1]$, this condition holds for all treated pages whenever

$$\kappa(\theta - \delta) \geq 0 \quad \text{and} \quad \rho > 0.$$

Under these parameter restrictions, the introduction of a Grokipedia entry weakly increases editing incentives for every treated page, with strict inequality for all pages with $s_i < 1$.

It follows immediately that

$$\mathbb{E}[E_i(1)] > \mathbb{E}[E_i(0)],$$

establishing that the introduction of Grokipedia increases expected Wikipedia editing activity on average. □

Appendix A.3 Proof of Proposition 2

For treated pages ($G_i = 1$), aggregate editing activity is

$$E_i(s_i) = F(\kappa \cdot \text{Views}_i + \rho(1 - s_i)).$$

Differentiating with respect to s_i yields

$$\frac{\partial E_i}{\partial s_i} = f(\kappa \cdot \text{Views}_i + \rho(1 - s_i)) \cdot (-\kappa\delta - \rho)\,.$$

Because $\kappa, \rho > 0$, $\delta \geq 0$, and $f(\cdot) > 0$, it follows that

$$\frac{\partial E_i}{\partial s_i} < 0.$$

Hence, among treated pages, editing activity is strictly decreasing in similarity. □

Appendix A.4 Proof of Proposition 3 (Irrelevance of Similarity for Control Pages)

For control pages ($G_i = 0$),

$$b_i = \kappa \cdot \bar{V}_i, \quad E_i = F(\kappa \cdot \bar{V}_i),$$

which does not depend on s_i. Therefore,

$$\frac{\partial E_i}{\partial s_i} = 0,$$

and similarity has no effect on editing activity for untreated pages. □

Appendix B Model Extension: Endogenous Similarity and Strategic Differentiation

This section extends the baseline model by allowing Wikipedia editing activity to endogenously affect the semantic similarity between Wikipedia and Grokipedia, and by introducing incentives for editors to deliberately differentiate Wikipedia content from AI-generated entries. We then show formally that this extension does not alter the main theoretical predictions.

Appendix B.1 Similarity Dynamics

Conditional on editing, an editor chooses a level of differentiation effort $d_i \geq 0$. Similarity evolves according to

$$s_i' = (1 - \lambda)s_i + \lambda \bar{s}_i - \eta d_i,$$

where $\bar{s}_i$ denotes the similarity that would arise absent strategic differentiation, $\lambda \in (0, 1)$ governs persistence, and $\eta > 0$ measures the effectiveness of differentiation effort. $\bar{s}_i$ reflects overlap due to shared facts and common source material.

Appendix B.2 Editing Benefits with Differentiation

With endogenous similarity, the gross benefit from editing page i is

$$B_i(d_i) = \kappa \cdot \text{Views}_i + \rho\Big(1 - s_i\Big) + \psi\delta\Big(1 - s_i'\Big),$$

where $\rho > 0$ captures an "inspiration" channel triggered by the pre-edit novelty of the AI entry relative to the existing Wikipedia text, and $\psi \geq 0$ captures a strategic differentiation motive: editors value reducing post-edit similarity in order to mitigate substitution when δ is large.

Page views are given by

$$\text{Views}_i = \bar{V}_i + G_i(\theta - \delta s_i'),$$

and editing costs are quadratic in differentiation effort.

Appendix B.3 Optimal Differentiation Choice

Let the cost of differentiation be

$$C(d_i) = \frac{c_d}{2} d_i^2, \qquad c_d > 0.$$

Under the extension, inspiration depends on the *pre-edit* novelty of the AI entry relative to Wikipedia, while substitution and strategic differentiation depend on the *post-edit* similarity s_i'. Specifically, conditional on editing, the editor chooses $d_i \geq 0$ to maximize

$$\max_{d_i \geq 0} \left\{ \kappa\Big(\bar{V}_i + G_i(\theta - \delta s_i')\Big) + G_i \rho(1 - s_i) + G_i \psi \delta (1 - s_i') - \frac{c_d}{2} d_i^2 \right\},$$

where similarity evolves as

$$s_i' = (1 - \lambda) s_i + \lambda \bar{s}_i - \eta d_i.$$

Lemma 2. ***(Endogenous Differentiation)*** *For treated pages ($G_i = 1$), the optimal differentiation effort is*

$$d_i^* = \frac{\eta}{c_d}(\kappa + \psi)\delta,$$

which is (weakly) increasing in the substitution parameter δ and strictly positive whenever $\delta > 0$. For untreated pages ($G_i = 0$), $d_i^* = 0$.

Proof. Substitute $s_i' = (1 - \lambda) s_i + \lambda \bar{s}_i - \eta d_i$ into the objective. The terms involving d_i are

$$\kappa\Big(- G_i \delta s_i' \Big) + G_i \psi \delta (1 - s_i') - \frac{c_d}{2} d_i^2 = \text{const} + G_i \eta d_i (\kappa + \psi)\delta - \frac{c_d}{2} d_i^2,$$

where $\rho(1 - s_i)$ does not depend on d_i because inspiration is triggered by the pre-edit similarity. For $G_i = 1$, the objective is strictly concave in d_i and the first-order condition is

$$\eta(\kappa + \psi)\delta - c_d d_i = 0,$$

which yields $d_i^* = \frac{\eta}{c_d}(\kappa + \psi)\delta$. For $G_i = 0$, d_i only increases costs and provides no benefits, so $d_i^* = 0$. □

Appendix B.4 Reduced-Form Net Benefit

Define the optimized net benefit

$$\tilde{b}_i \equiv \max_{d_i \geq 0} b_i(d_i),$$

where $b_i(d_i)$ is the editor's payoff inside the braces above.

Lemma 3. *For treated pages ($G_i = 1$), the optimized net benefit is*

$$\tilde{b}_i = \kappa\Big(\bar{V}_i + \theta - \delta\Big[(1-\lambda)s_i + \lambda\bar{s}_i\Big]\Big) + \rho(1-s_i) + \psi\delta\Big(1 - \Big[(1-\lambda)s_i + \lambda\bar{s}_i\Big]\Big) + \frac{\eta^2}{2c_d}\Big((\kappa+\psi)\delta\Big)^2.$$

For untreated pages ($G_i = 0$), $\tilde{b}_i = \kappa\bar{V}_i$.

Proof. Let $s_i^0 \equiv (1-\lambda)s_i + \lambda\bar{s}_i$ denote post-edit similarity absent differentiation ($d_i = 0$). Substituting d_i^* from Lemma 2 implies $s_i' = s_i^0 - \eta d_i^*$. Plugging d_i^* into the objective yields the stated expression, where the last term is the standard quadratic maximization gain:

$$\max_{d_i \geq 0}\left\{\eta d_i(\kappa+\psi)\delta - \frac{c_d}{2}d_i^2\right\} = \frac{\eta^2}{2c_d}\Big((\kappa+\psi)\delta\Big)^2.$$

For $G_i = 0$, the objective does not contain s_i' and d_i, so $\tilde{b}_i = \kappa\bar{V}_i$. □

Aggregate editing activity remains

$$E_i = F(\tilde{b}_i),$$

where $F(\cdot)$ is the CDF of editing costs.

Appendix B.5 Invariance of the Main Propositions

We now show that Propositions 1–3 continue to hold.

Proposition 1. From Lemma 3, for treated pages,

$$\tilde{b}_i(1) - \tilde{b}_i(0) = \kappa\Big(\theta - \delta s_i^0\Big) + \rho(1-s_i) + \psi\delta(1-s_i^0) + \frac{\eta^2}{2c_d}\Big((\kappa+\psi)\delta\Big)^2, \quad s_i^0 = (1-\lambda)s_i + \lambda\bar{s}_i.$$

Since $s_i^0 \in [0,1]$, the condition $\theta \geq \delta$ implies $\kappa(\theta - \delta s_i^0) \geq \kappa(\theta - \delta) \geq 0$. Moreover, $\rho(1 - s_i) \geq 0$ with strict inequality whenever $s_i < 1$, and the remaining terms are nonnegative. Therefore $\tilde{b}_i(1) \geq \tilde{b}_i(0)$ pointwise, with strict inequality on a set of positive measure. Because $F(\cdot)$ is strictly increasing, it follows that $E_i(1) \geq E_i(0)$ and Proposition 1 holds.

Proposition 2. For treated pages, differentiating Lemma 3 with respect to s_i yields

$$\frac{\partial \tilde{b}_i}{\partial s_i} = -\rho - (1-\lambda)\delta(\kappa + \psi) < 0,$$

since $\rho > 0$, $\kappa > 0$, $\delta \geq 0$, $\psi \geq 0$, and $\lambda \in (0,1)$. Since $f(\cdot) > 0$,

$$\frac{\partial E_i}{\partial s_i} = f(\tilde{b}_i)\frac{\partial \tilde{b}_i}{\partial s_i} < 0,$$

establishing Proposition 2.

Proposition 3. For untreated pages, $\tilde{b}_i = \kappa \bar{V}_i$ does not depend on s_i, implying

$$\frac{\partial E_i}{\partial s_i} = 0.$$

Thus Proposition 3 continues to hold. □

Appendix C Detailed Matching Summary Statistics

Table A1 reports treated and matched-control means for each of the eight pre-treatment weeks, together with mean gaps, standard errors and standardized differences. Standardization uses the complete eligible population within each field. Each use of a control receives the same weight as the treated page to which it is assigned. The exact edit-profile restriction requires identical total-edit counts in every pre-treatment week, so the total-edit gaps and their standard errors are mechanically zero.

The largest absolute standardized weekly mean difference (SMD) across views, total edits and minor edits is 0.043, and the corresponding maximum slope difference is 0.025. Most values are substantially smaller; an absolute SMD below 0.1 is a commonly used benchmark for good mean balance (Austin, 2009). With the pooled matched-group standard deviation as denominator, the maxima are 0.091 and 0.062, respectively. Exact total-edit balance should not be interpreted as an independent validation of the identifying assumption.

Table A2 reports the three historical matching covariates. Biography length remains imbalanced in several fields, with a maximum SMD of 0.469 in Academia. These diagnostics delimit the recent-activity balance claim; matching does not equate every historical characteristic.

Table A1: Eight-Week Pre-Treatment Outcome Balance

Field	Week	Treated	Control	Gap	SE	SMD	Pairs
Page views							
Academia	-8	315.753	285.405	30.348	5.139	0.028	4,808
Academia	-7	303.996	281.830	22.166	4.219	0.024	4,808
Academia	-6	303.141	309.649	-6.507	4.585	-0.003	4,808
Academia	-5	309.261	325.756	-16.495	5.940	-0.015	4,808
Academia	-4	314.222	312.017	2.205	4.807	0.000	4,808
Academia	-3	327.674	337.512	-9.838	5.467	-0.005	4,808
Academia	-2	315.833	319.735	-3.902	5.161	-0.004	4,808
Academia	-1	316.360	304.677	11.683	4.124	0.012	4,808
Culture	-8	493.898	479.298	14.600	3.387	0.003	32,690
Culture	-7	454.975	459.796	-4.821	5.085	-0.001	32,690
Culture	-6	458.600	450.013	8.586	2.715	0.001	32,690
Culture	-5	462.385	447.982	14.403	3.373	0.003	32,690
Culture	-4	466.923	458.216	8.707	3.073	0.002	32,690
Culture	-3	468.326	487.386	-19.060	5.630	-0.002	32,690
Culture	-2	467.651	457.423	10.229	2.891	0.001	32,690
Culture	-1	483.278	480.384	2.894	4.406	0.001	32,690
Leaders	-8	446.141	426.155	19.987	4.986	0.005	12,880
Leaders	-7	412.729	413.452	-0.723	4.045	-0.000	12,880
Leaders	-6	421.033	401.891	19.142	3.661	0.007	12,880

Field	Week	Treated	Control	Gap	SE	SMD	Pairs
Leaders	-5	427.477	444.626	-17.150	4.535	-0.006	12,880
Leaders	-4	423.516	460.380	-36.864	5.028	-0.012	12,880
Leaders	-3	431.097	420.585	10.512	4.428	0.003	12,880
Leaders	-2	430.170	419.176	10.994	4.010	0.004	12,880
Leaders	-1	443.468	419.975	23.493	3.478	0.010	12,880
Politics	-8	365.238	335.320	29.918	3.852	0.011	8,461
Politics	-7	347.800	356.043	-8.242	5.037	-0.002	8,461
Politics	-6	341.304	348.781	-7.477	4.920	-0.003	8,461
Politics	-5	343.824	331.633	12.191	3.087	0.004	8,461
Politics	-4	340.990	347.951	-6.961	4.871	-0.002	8,461
Politics	-3	347.545	352.457	-4.911	4.032	-0.001	8,461
Politics	-2	348.174	340.074	8.100	3.528	0.003	8,461
Politics	-1	358.884	351.448	7.436	3.668	0.003	8,461
Sports	-8	329.343	326.231	3.112	4.743	0.001	22,070
Sports	-7	294.597	316.640	-22.043	3.722	-0.000	22,070
Sports	-6	295.805	312.027	-16.222	3.512	-0.003	22,070
Sports	-5	297.204	309.539	-12.335	3.939	-0.004	22,070
Sports	-4	297.617	285.564	12.053	2.924	0.006	22,070
Sports	-3	308.261	301.037	7.223	3.981	0.004	22,070
Sports	-2	310.848	286.417	24.430	3.130	0.012	22,070
Sports	-1	310.832	283.183	27.649	3.924	0.009	22,070
Total edits							
Academia	-8	0.056	0.056	0.000	0.000	0.000	4,808
Academia	-7	0.058	0.058	0.000	0.000	0.000	4,808
Academia	-6	0.064	0.064	0.000	0.000	0.000	4,808
Academia	-5	0.075	0.075	0.000	0.000	0.000	4,808
Academia	-4	0.102	0.102	0.000	0.000	0.000	4,808
Academia	-3	0.052	0.052	0.000	0.000	0.000	4,808
Academia	-2	0.061	0.061	0.000	0.000	0.000	4,808
Academia	-1	0.051	0.051	0.000	0.000	0.000	4,808
Culture	-8	0.088	0.088	0.000	0.000	0.000	32,690
Culture	-7	0.074	0.074	0.000	0.000	0.000	32,690
Culture	-6	0.082	0.082	0.000	0.000	0.000	32,690
Culture	-5	0.053	0.053	0.000	0.000	0.000	32,690
Culture	-4	0.063	0.063	0.000	0.000	0.000	32,690
Culture	-3	0.074	0.074	0.000	0.000	0.000	32,690
Culture	-2	0.070	0.070	0.000	0.000	0.000	32,690
Culture	-1	0.074	0.074	0.000	0.000	0.000	32,690
Leaders	-8	0.051	0.051	0.000	0.000	0.000	12,880
Leaders	-7	0.054	0.054	0.000	0.000	0.000	12,880
Leaders	-6	0.059	0.059	0.000	0.000	0.000	12,880
Leaders	-5	0.055	0.055	0.000	0.000	0.000	12,880
Leaders	-4	0.056	0.056	0.000	0.000	0.000	12,880
Leaders	-3	0.057	0.057	0.000	0.000	0.000	12,880
Leaders	-2	0.045	0.045	0.000	0.000	0.000	12,880
Leaders	-1	0.087	0.087	0.000	0.000	0.000	12,880
Politics	-8	0.091	0.091	0.000	0.000	0.000	8,461
Politics	-7	0.068	0.068	0.000	0.000	0.000	8,461
Politics	-6	0.114	0.114	0.000	0.000	0.000	8,461
Politics	-5	0.089	0.089	0.000	0.000	0.000	8,461
Politics	-4	0.079	0.079	0.000	0.000	0.000	8,461
Politics	-3	0.051	0.051	0.000	0.000	0.000	8,461
Politics	-2	0.073	0.073	0.000	0.000	0.000	8,461
Politics	-1	0.077	0.077	0.000	0.000	0.000	8,461
Sports	-8	0.131	0.131	0.000	0.000	0.000	22,070
Sports	-7	0.074	0.074	0.000	0.000	0.000	22,070
Sports	-6	0.126	0.126	0.000	0.000	0.000	22,070
Sports	-5	0.122	0.122	0.000	0.000	0.000	22,070
Sports	-4	0.085	0.085	0.000	0.000	0.000	22,070
Sports	-3	0.089	0.089	0.000	0.000	0.000	22,070
Sports	-2	0.118	0.118	0.000	0.000	0.000	22,070

Field	Week	Treated	Control	Gap	SE	SMD	Pairs
Sports	-1	0.105	0.105	0.000	0.000	0.000	22,070
Minor edits							
Academia	-8	0.019	0.015	0.004	0.002	0.020	4,808
Academia	-7	0.019	0.012	0.007	0.002	0.035	4,808
Academia	-6	0.011	0.012	-0.001	0.002	-0.004	4,808
Academia	-5	0.028	0.028	0.000	0.003	0.001	4,808
Academia	-4	0.058	0.050	0.008	0.003	0.024	4,808
Academia	-3	0.015	0.026	-0.011	0.002	-0.040	4,808
Academia	-2	0.022	0.019	0.003	0.002	0.012	4,808
Academia	-1	0.018	0.007	0.010	0.002	0.043	4,808
Culture	-8	0.026	0.025	0.001	0.001	0.003	32,690
Culture	-7	0.015	0.016	-0.001	0.001	-0.002	32,690
Culture	-6	0.011	0.010	0.001	0.001	0.003	32,690
Culture	-5	0.012	0.012	0.001	0.001	0.002	32,690
Culture	-4	0.016	0.015	0.000	0.001	0.001	32,690
Culture	-3	0.022	0.024	-0.002	0.001	-0.007	32,690
Culture	-2	0.023	0.021	0.002	0.001	0.006	32,690
Culture	-1	0.025	0.022	0.003	0.001	0.008	32,690
Leaders	-8	0.016	0.011	0.005	0.001	0.024	12,880
Leaders	-7	0.011	0.012	-0.001	0.001	-0.006	12,880
Leaders	-6	0.008	0.010	-0.001	0.001	-0.009	12,880
Leaders	-5	0.017	0.020	-0.003	0.001	-0.015	12,880
Leaders	-4	0.015	0.013	0.002	0.001	0.008	12,880
Leaders	-3	0.017	0.021	-0.004	0.002	-0.018	12,880
Leaders	-2	0.014	0.012	0.003	0.001	0.013	12,880
Leaders	-1	0.044	0.042	0.002	0.002	0.007	12,880
Politics	-8	0.021	0.030	-0.009	0.002	-0.040	8,461
Politics	-7	0.018	0.015	0.004	0.002	0.017	8,461
Politics	-6	0.013	0.016	-0.003	0.002	-0.017	8,461
Politics	-5	0.031	0.034	-0.003	0.002	-0.010	8,461
Politics	-4	0.021	0.024	-0.004	0.002	-0.013	8,461
Politics	-3	0.016	0.015	0.001	0.002	0.003	8,461
Politics	-2	0.023	0.019	0.004	0.002	0.016	8,461
Politics	-1	0.021	0.017	0.004	0.002	0.011	8,461
Sports	-8	0.016	0.018	-0.001	0.001	-0.006	22,070
Sports	-7	0.017	0.019	-0.001	0.001	-0.002	22,070
Sports	-6	0.020	0.026	-0.006	0.001	-0.021	22,070
Sports	-5	0.059	0.053	0.006	0.002	0.022	22,070
Sports	-4	0.029	0.035	-0.005	0.001	-0.026	22,070
Sports	-3	0.030	0.035	-0.005	0.001	-0.021	22,070
Sports	-2	0.058	0.066	-0.008	0.002	-0.017	22,070
Sports	-1	0.045	0.040	0.005	0.002	0.018	22,070

Notes: Weeks −8 through −1 are September 1–October 26, 2025. SMD divides the mean gap by the standard deviation in the complete eligible field population. Control means weight each use of a control. Gap standard errors use pair-level differences. All three historical covariates remain in matching. Equality of the eight weekly total-edit counts is imposed by matching, rather than independent evidence of parallel trends.

Table A2: Historical Covariate Balance in the Matched Sample

Field	Historical covariate	Treated	Control	SMD
Academia	60-day edits	0.473	0.439	0.012
Academia	Biography word count	12,243.6	9,169.8	0.469
Academia	2015–2018 readership	128,886.5	109,259.0	0.035
Culture	60-day edits	0.516	0.497	0.004
Culture	Biography word count	7,656.1	6,547.9	0.147
Culture	2015–2018 readership	228,298.7	244,942.1	-0.014
Leaders	60-day edits	0.401	0.363	0.011
Leaders	Biography word count	11,035.9	10,103.9	0.120
Leaders	2015–2018 readership	155,522.5	185,656.3	-0.034
Politics	60-day edits	0.554	0.495	0.015
Politics	Biography word count	9,447.3	7,005.3	0.355
Politics	2015–2018 readership	132,157.8	100,951.5	0.048
Sports	60-day edits	0.696	0.656	0.008
Sports	Biography word count	11,590.3	8,760.3	0.387
Sports	2015–2018 readership	151,212.2	102,308.9	0.100

SMD is the treated-minus-control mean divided by the sample standard deviation in the complete eligible field population. Controls are weighted by their use in matched pairs. Historical covariates enter in their original units.

Appendix D Citizenship and Birth-Year Matching

This appendix examines whether comparisons within nationality and birth cohort support the main average effects. We add exact matching on primary citizenship and birth year as a standardized distance coordinate with unit multiplier. The procedure retains the preferred weighted activity history and historical covariates, exact eight-week total-edit profile, view caliper, 250-candidate search and reuse cap of ten. Missing citizenship or birth year excludes a page. Standardization uses the demographically eligible population within each field. The eight-pre/eight-post timing is unchanged.

The match retains 53,249 treated pages. The smaller sample reflects missing demographic information and narrower comparison groups. Table A3 reports 406.02 additional views (standard error 33.63), 0.294 total edits (0.019) and 0.080 minor edits (0.005). All three are positive and significant at the 1 percent level. These estimates support a positive activity response under the additional demographic restrictions, although the retained treated sample differs from that in Table 2.

Table A3: Pooled Eight-Week DiD: Citizenship and Birth-Year Matching

	Views	Total edits	Minor edits
DiD	406.02***	0.294***	0.080***
	(33.63)	(0.019)	(0.005)
Matched pairs	53,249	53,249	53,249

Standard errors in parentheses. Matching uses eight pre-treatment weeks, September 1–October 26, 2025, and the three historical covariates. Post-treatment weeks cover October 27–December 21, 2025. DiD coefficients and standard errors are reported in eight-week total-count units: divide both by eight for the difference between average weekly post- and pre-treatment gaps. One control per treated page, with at most ten uses per control. DiD standard errors use pair-level differences and are not adjusted for reused controls. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$. Matching uses weighted standardized Euclidean distance, exact eight-week total-edit profiles within field, and a 0.10 log-mean-view caliper. This robustness adds exact primary citizenship and standardized birth year, excluding missing demographic values.

Appendix E Similarity and Wikipedia Page Views

We examine whether textual overlap predicts changes in Wikipedia traffic after launch. We apply the regressions in Section 6.4 to each treated page's view DiD relative to its baseline matched control, in eight-week total-count units. The specifications successively include similarity alone, field indicators, and historical word count, gender and age.

Table A4: Similarity and Eight-Week Matched-Pair View Changes

	Views (1)	Views (2)	Views (3)
Similarity score	592.4*	530.8	638.5*
	(320.1)	(324.0)	(345.4)
Field indicators	No	Yes	Yes
Page controls	No	No	Yes
Observations	80,687	80,687	75,450
R^2	0.000	0.000	0.000

Standard errors in parentheses. Outcomes are individual matched-pair DiD changes in eight-week total-count units. Both periods contain eight weeks: September 1–October 26 and October 27–December 21, 2025. Conventional OLS inference. Page controls are historical word count, gender and age. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

All three specifications in Table A4 yield positive coefficients. With full controls, a 0.10 increase in similarity is associated with 63.9 additional views, or about 2.0% of the treated sample's eight-week baseline (coefficient 638.55; standard error 345.39; $p = 0.064$).

Appendix B shows how subsequent differentiation can also affect the relationship between similarity and readership.

Appendix F Textual Convergence by Field

This appendix estimates the controlled convergence regression in column (3) of Table 5 separately within each occupational field. The dependent variable is $\Delta S_i = S_{i8} - S_{i0}$, and initial gap is $g_i = 1 - S_{i0}$. Each regression includes historical word count, gender and age, with the same balanced nine-snapshot sample construction and HC1 inference as the pooled specification. Field indicators are omitted because each sample contains only one field; control coefficients may vary across fields. These regressions contain no post-launch editing variable or editing interaction.

Table A5: Initial Textual Gap and Eight-Week Similarity Change by Field

	Academia	Culture	Leaders	Politics	Sports
Initial gap	0.016400***	0.006711***	0.008324***	0.008567***	0.003458***
	(0.003111)	(0.000740)	(0.001467)	(0.002135)	(0.000873)
Page controls	Yes	Yes	Yes	Yes	Yes
Observations	4,506	29,738	11,527	7,873	21,777
R^2	0.012	0.004	0.005	0.005	0.002

Standard errors in parentheses. Each column estimates the field-specific counterpart of Table 5, column (3), using balanced nine-snapshot pages with available controls. The outcome is week-eight minus pre-launch similarity against fixed October 27 Grokipedia text. Initial gap is one minus pre-launch similarity. Controls are historical word count, gender and age. Field indicators are omitted within each field. No post-editing variable or editing interaction is included. HC1 robust inference uses residual-degrees-of-freedom t statistics. Positive coefficients describe greater subsequent similarity change among initially more distant pages. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

The initial-gap coefficient is positive in all five fields. Academia has the largest point estimate, 0.016400, while the estimates for Politics and Leaders are 0.008567 and 0.008324, respectively. Sports has the smallest point estimate, 0.003458. The 95% confidence intervals are $[0.010302, 0.022499]$ for Academia and $[0.001747, 0.005169]$ for Sports. These two intervals do not overlap, although several other field intervals do, including those for Leaders and Politics. A pooled regression with initial-gap-by-field interactions rejects equal slopes across fields (HC1 robust $F(4, 75, 406) = 6.42$, $p = 3.65 \times 10^{-5}$). This complementary test uses common page-control coefficients, whereas the separate regressions allow those coefficients to vary. The results indicate a common positive sign with differences in magnitude.